Uncertainty Quantification of Hyperspectral Image Denoising Frameworks Based on Sliding-Window Low-Rank Matrix Approximation

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***Abstract*— Sliding-window-based low-rank matrix approximation (LRMA) is a technique widely used in hyperspectral images (HSIs) denoising or completion. However, the uncertainty quantification of the restored HSI has not been addressed to date. Accurate uncertainty quantification of the denoised HSI facilitates applications such as multisource or multiscale data fusion, data assimilation, and product uncertainty quantification since these applications require an accurate approach to describe the statistical distributions of the input data. Therefore, we propose a prior-free closed-form element-wise uncertainty quantification method for LRMA-based HSI restoration. Our closed-form algorithm overcomes the difficulty of handling uncertainty in HSI patch mixing caused by the sliding-window strategy used in the conventional LRMA process. The proposed approach only requires the uncertainty of the observed HSI and provides the uncertainty result relatively rapidly and with similar computational complexity as the LRMA technique. We conduct extensive experiments to validate the estimation accuracy of the proposed closed-form uncertainty approach. The method is robust to at least 10% random impulse noise at the cost of 10%–20% of additional processing time compared to the LRMA. The experiments indicate that the proposed closed-form uncertainty quantification method is more applicable to real-world applications than the baseline Monte Carlo test, which is computationally expensive.**

***Index Terms*— Hyperspectral images (HSIs), low-rank matrix approximation (LRMA), uncertainty quantification.**

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I. Introduction

DATA uncertainty, which refers to the lack of sureness about the data [1], is an essential factor to consider when utilizing data and derived products [2]. When the data uncertainty is known, users can determine whether a product is suitable for specific applications or can balance the contributions of different data sources. For example, in [3], the minimum detectable deformation can be estimated based on the uncertainty measures in a point cloud so that engineers can develop appropriate project plans, including sensor selection and set-up schemes [4]. In multiple data fusion tasks, the label uncertainties of hyperspectral images (HSIs) and point clouds are often considered to achieve high fusion accuracy [5].

In theory, uncertainty can be classified as aleatoric uncertainty and epistemic uncertainty [6]. Aleatoric uncertainty is caused by noise in the data and the processing steps. In contrast, epistemic uncertainty is introduced into the observation models, and data post-processing steps [1], [2], [4], [7], [8]. In the absence of ground truth data to measure the model performance, the epistemic uncertainty is ignored, and the output uncertainty is defined as the aleatoric uncertainty due to the propagation of an arbitrary model. Most studies have defined the probabilistic density function of the noise as a mixture of the normal distribution and the random distribution. Assuming that random noise is significantly low, the uncertainty can be described by the Gaussian distribution. In HSI restoration, symmetric uncertainty based on mutual information was proposed in [9] and was used in HSI classifications. In HSI unmixing, the end member uncertainty of unmixing methods based on the normal compositional model is defined as the covariance matrix, which is used to predict the unmixing errors without knowing the ground truth [10]. Uncertainty has been described differently in different studies. Dehghan and Ghassemian [1] proposed an entropy-based uncertainty measure for evaluating the classification results of multispectral remote sensing images. The calculated entropy, which was defined as the pixel-wise classification uncertainty, had a linear relation with the accuracy of the classification results. In [11], the uncertainty of 3-D point clouds was defined as the measurement repeatability, which was represented by the median and the two percentiles (5% and 95%) related to repeated measurements.

This article focuses on the uncertainty quantification in HSI denoising. HSIs consisting of hundreds or thousands of spectral bands have been widely used in various applications (e.g.,

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land cover mapping, pollution monitoring, and crop yield estimation) due to their rich spectral information compared to the spatial information. In HSI data processing, image denoising is regarded as an important preprocessing step for advanced applications and has been researched intensively in the past decades [12]. According to [13], existing HSI denoising methods can be categorized into three groups: filter-based methods, model optimization methods, and deep learning methods. Filter-based methods try to remove noise using various filters. Classical methods in this category are based on nonlocal filters [14]–[17]. Model optimization methods can be further divided into total variation methods [18], sparse representation-based [19], and sliding-window low-rank matrix-based methods [20]. The major advantages of this group of denoising methods are utilizing image priors and considering spatial and spectral features simultaneously. These methods are often combined to achieve better performance [21], [22]. The last group is based on deep learning frameworks, which have attracted increasing attention in the past three years [23]. A recent deep learning-based framework was proposed in [13] to remove hybrid noise in a spatial-spectral framework.

HSI denoising based on the sliding-window low-rank matrix approximation (LRMA) framework has become a state-of-the-art technique in the community [20], [24]. These methods are model optimization methods aimed at exploiting the low-dimensional structure in the high-dimensional data space. LRMA-based studies [20], [21], [25] assume that the noise-free HSI can be represented by a low-rank matrix. Thus, HSI denoising can be regarded as a classical sliding-window low-rank matrix recovery problem. A representative HSI denoising framework based on LRMA is described in [20], where a clean image, impulse noise, and Gaussian noise were formulated separately in the objective function, which was minimized under a low-rank assumption. Many variants of LRMA-based HSI denoising methods have been proposed in the past five years [21], [24], [26]–[28]. For example, [24] proposed an LRMA-based framework containing a total-variation regularization term to preserve spatial information. The spatial-spectral structure in HSI denoising was also exploited by replacing the sliding-window low-rank matrix analysis with low-rank tensor approximation [28]. In general, LRMA-based HSI denoising methods are state-of-the-art frameworks that have been widely used in this area.

Despite the progress in LRMA-based HSI denoising, no studies investigated the uncertainty quantification of denoised HSI to date. Some works aimed to determine the noise of the raw observations [29], [30], but the uncertainty of denoised HSI has not yet been addressed, although it is critical in subsequent applications, as discussed above. However, the quantification of uncertainties of denoised HSIs based on LRMA is difficult due to the nonlinear discontinuous functions in LRMA models (e.g., nuclear norm), which hinders the deduction of the closed-form uncertainty propagation formulations. Recently, the study [31] proposed an optimal uncertainty quantification and inference method for noisy matrix completion. Inspired by this work, we expect that it is possible to quantify the uncertainties of outputs from LRMA-based HSI denoising frameworks.

A sliding-window strategy is always used in HSI denoising to handle the extensive data volume of HSI [20], [24]; the denoised element values are obtained by averaging the values from several traversed patches. The mixed values from multiple patches pose difficulty in quantifying the uncertainty of the estimations. Unlike the conventional perspective of multiple observations, which obey the same independent and identical distribution (i.i.d.), multiple overlapping sliding windows are interconnected statistically. Thus, the covariance of the overlapping window should also be addressed. To tackle these problems, we introduce and improve the method described in [32] to quantify the uncertainty of the low-rank-based HSI approach. Note that we assume the input HSI is a mixture of Gaussian noise and random noise. Previous studies [16], [21], [30] already proved that the Poisson noise was the primary concern in real HSI data but could be approximated by additive Gaussian noise. Based on these results, Gaussian noise with mean and covariance is suitable to establish the uncertainty propagation model.

This research does not aim at providing a new LRMA formulation. Instead, we propose a general uncertainty quantification method for the existing LRMA approach. The main contributions of this article are threefold,

1) To the best of our knowledge, the uncertainty quantification of the denoised HSI based on LRMA is presented for the first time in this article. We provide a closed-form time-efficient uncertainty propagation model to predict the element-wise uncertainty of the denoised HSI without any ground truth data.
2) To solve the "uncertainty mixing" problem caused by the sliding-window strategy, we deduce a weighted average uncertainty formulation, which proves effective based on extensive experiments.
3) The proposed uncertainty estimation approach is independent of the choice of the LRMA algorithm. It is based on the global minimum assumption of the LRMA solver; thus, a more accurate solver leads to better uncertainty description.

The rest of this article is organized as follows. In Section II, we briefly summarize the canonical formulation of LRMA-based HSI denoising. Section III presents the proposed algorithm for the uncertainty propagation in a classical LRMA-based HSI denoising framework. Section IV shows experiments on both simulated and real HSI data. The discussion of the parameter settings and limitations is also presented in Section IV. Section V summarizes this article.

II. LRMA Algorithm for LRMA HSI Denoising

We cite [20] as a typical LRMA approach and discuss the uncertainty evaluation based on the LRMA approach. Section II is a brief formulation of the routine LRMA method [20]. The results provided in Section II have been strictly tested and validated in different LRMA studies. Section III describes the proposed uncertainty estimation method for an LRMA approach, as described in [20]. We emphasize that this work DOES NOT aim at proposing another LRMA formulation. The goal of this approach is to quantify the uncertainty of LRMA denoised images in a closed-form manner.

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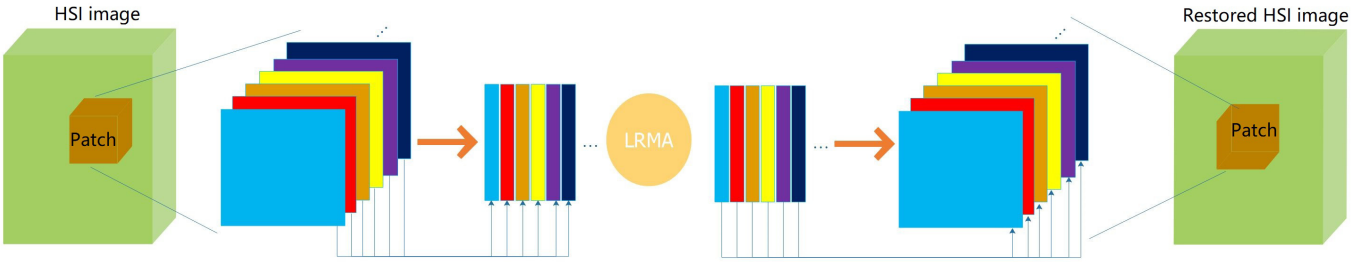


Fig. 1. Illustration of the routine process of HSI denoising. The 3-D matrices at each end represent the raw and denoised HSI, respectively. The sliding window operates on each patch of the matrix. The selected 3-D patch matrix is converted to a 2-D matrix by concatenating each band of the 2-D matrix as a 1-D vector. The permutated 2-D matrix goes through the LRMA process to generate the denoised matrix. Symmetrically, the denoised 2-D matrix is inversely permutated to the 3-D patch matrix and constitutes the final restored HSI in 3-D.

A. Revisiting the General HSI Restoration Process

Fig. 1 demonstrates the routine process of the LMRA-based HSI restoration. The HSI is processed with a 3-D sliding window to reduce the computational burden. For each 3-D patch in the window, all the bands are vectorized and stacked into the 2-D matrix. The permutated 2-D matrix goes through the routine LMRA process, and the result is repermutated to recover the 3-D denoised patch. Finally, all the patches are summarized and averaged to create the restored HSI.

An HSI acquired by a sensor is typically modeled as a 3-D image composed of Gaussian noise and random noise

$$\mathbf{W} = \mathbf{W}^{\star} + \mathbf{G} + \mathbf{N} \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{M\times N\times P}$ is the image acquired from the hyperspectral sensor with the size $M \times N$ and P channels, $\mathbf{W}^{\star} \in \mathbb{R}^{M\times N\times P}$ is the noise-free image denoted as the ground truth, $\mathbf{G} \in \mathbb{R}^{M\times N\times P}$ is the element-wise i.i.d. Gaussian noise and $\mathbf{N} \in \mathbb{R}^{M\times N\times P}$ is the sparse and random noise (impulse signal). The random noise $\mathbf{N}$ is normally considered negligible with regard to $\mathbf{G}$, and the data quality quantification process is inferring the statistical distribution of $\mathbf{G}$. In practice, the ground truth HSI $\mathbf{W}^{\star}$ cannot be obtained. Thus, we denote $\hat{\mathbf{W}}$ as the estimated denoised HSI calculated from studies based on the low-rank assumption [21], [22].

The goal of HSI denoising is to recover the denoised $\hat{\mathbf{W}}$ with the 3-D HSI noisy observation $\mathbf{W}$ and the exact rank prior. Assuming that the random noise $\mathbf{N}$ is negligible and the residual follows the i.i.d. zero-mean Gaussian distribution, the restored image $\hat{\mathbf{W}}$ is recovered with the following formulation:

$$\begin{aligned} &\min_{\hat{\mathbf{W}}} \mu \operatorname{rank}\left(\hat{\mathbf{W}}^{\sharp}\right) + \|\mathbf{G}\|_F^2, \ \text{s.t. } \hat{\mathbf{W}} = \mathbf{W} + \mathbf{G} \\ &\text{s.t. } \hat{\mathbf{W}}^{\sharp} = g\left(\hat{\mathbf{W}}\right) \end{aligned} \tag{2}$$

where μ is the hyperparameter balancing the two terms. rank$(\cdot)$ constraints the rank of the input matrix to be no greater than the given rank. $||\cdot||_F$ is the Frobenius norm that minimizes the residuals in Gaussian distribution. $g(\cdot)$ permutates the 3-D HSI to a 2-D matrix by vectorizing and concatenating to a general 2-D matrix (Fig. 1). The low-rank formulation was originally proposed by [33] and was called "robust principal component analysis" (RPCA). The work in [20] introduced RPCA into LRMA scenario.

B. Sliding-Window in the HSI Restoration Process

HSI restoration differs greatly from conventional image processing due to the data size. The extremely large permutated 2-D matrix hinders the iterative regular singular value decomposition (SVD) process, which is essential in solving (2). The HSI is segmented into small patches (denoted as the overlapped sliding window) and solved in parallel to overcome the heavy computational burden in minimizing the objective function due to the huge HSI size. Previous studies regarded the sliding-window strategy as an engineering compromise and ignored the formal mathematical presentation. However, we realize that the unique sliding-window structure significantly impairs the uncertainty quantification. By following [20], we obtain a sliding-window version that minimizes all patches in (2):

$$\begin{aligned} &\min_{\hat{\mathbf{W}}^{(ijk)}} \mu \operatorname{rank}\left(\hat{\mathbf{W}}^{\sharp(ijk)}\right) + \frac{1}{2}\|\mathbf{W}^{(ijk)} - \hat{\mathbf{W}}^{(ijk)}\|_F^2 \\ &\text{s.t. } \hat{\mathbf{W}}^{\sharp(ijk)} = g\left(\hat{\mathbf{W}}^{(ijk)}\right) \\ &\qquad (i,j,k) \in \Omega \end{aligned} \tag{3}$$

where the matrices $\mathbf{W}^{(ijk)}$, $\hat{\mathbf{W}}^{(ijk)}$ and $\hat{\mathbf{W}}^{\sharp(ijk)} \in \mathbb{R}^{J\times J}$ are the patch indexed in (ijk) with regard to $\mathbf{W}$, $\hat{\mathbf{W}}$ and $\hat{\mathbf{W}}^{\sharp}$. J is the size of the permutated matrix. $\Omega \subseteq \{1,\ldots,M'\} \times \{1,\ldots,N'\} \times \{1,\ldots,P'\}$ is the number of patches in the HSI. After all the patches are optimized, the optimal $\hat{\mathbf{W}}$ is restored by averaging the overlapped patches $\hat{\mathbf{W}}^{(ijk)}$ element-wisely. By ignoring the random noise, we obtain the observation corrupted with the element-wise Gaussian noise

$$\mathbf{W}^{(ijk)} \approx \mathbf{W}^{\star(ijk)} + \mathbf{G}^{(ijk)}, \quad \mathbf{G}^{(ijk)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma_0^2\right) \tag{4}$$

where σ_0^2 is the variance of the noise obtained from the image or estimated by algorithms such as described in [25]. $\mathbf{G}^{(ijk)}$ is the Gaussian noise in patch (ijk). In previous studies, the choice of rank was often estimated by manual-tuning strategies based on the differences between the reconstructed signal and the original one [21].

The idea of (3) is to convert the 3-D patch into a typical 2-D small-size matrix so that the conventional low-rank factorization algorithm can be executed with limited computational resources. The equation is often solved with a Lagrange multiplier or an augmented Lagrange multiplier (ALM); off-the-shelf low-rank solvers can be used. In this work, we choose

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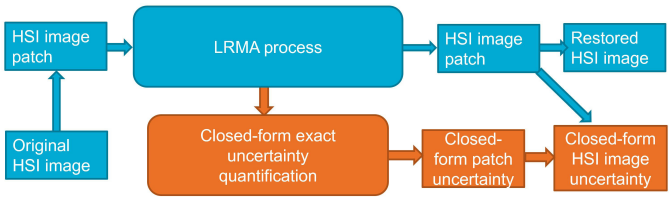


Fig. 2. General framework of the proposed approach. The blue boxes represent the typical LRMA process ([20] as an example). In addition to the LRMA HSI denoising, our work aims at providing an extra module to quantify the uncertainty of the output and the restored HSI using the LRMA algorithm.

the GoDec algorithm proposed in [34] to be consistent with the LRMA method [20]. However, it should be emphasized that the proposed uncertainty quantification method is independent of the solver since it only assumes that the estimation of the LRMA is close to the minimum. The better the performance of the LRMA solver, the more accurate the confidence interval of our method describes.

After all small overlapped patches have gone through the process of (3), the optimized denoised patches are summed to restore the HSI

$$\hat{\mathbf{W}} = \left[\sum_{i,j,k \in \Omega} f_p\left(\hat{\mathbf{W}}^{(ijk)}\right) \right] \oslash \mathbf{Q} \tag{5}$$

where $f_p(\cdot)$ is the padding function to convert the patch to the same size of $\hat{\mathbf{W}}$ by padding the rest of the elements with 0. The matrix $\mathbf{Q} \in \mathbb{R}^{M \times N \times P}$ counts the number of the patches valid on each voxel. $\oslash$ is the Hadamard division operator.

III. Uncertainty Quantification Method for LRMA-Based HSI Denoising

Based on the above-mentioned formulation, we propose a closed-form element-wise uncertainty quantification method that can handle the sliding-window LRMA-based HSI denoising.

A. General Framework

Fig. 2 illustrates the general framework and the relationship with the routine LRMA approach. In the LRMA process, the temporary variable, i.e., the 2-D sliding-window patch, is utilized to provide the uncertainty to the HSI patch based on our closed-form formulation. After the HSI patch and its associated closed-form uncertainty have been generated, the routine LRMA approach averages them to yield the restored HSI. We also provide a closed-form uncertainty propagation to quantify the final restored HSI. As shown in Fig. 2, in all processes, the proposed approach is independent of the choice of the low-rank solver. The method can also be easily integrated into the derivations of the LRMA process. In addition, since our method is a closed-form solution and only utilizes temporary variables, the processing time is relatively short compared to the routine LRMA process.

One major contribution of this work is that it provides an element-wise closed-form solution to quantify the uncertainty in a time-efficient manner. For the restored HSI $\hat{\mathbf{W}}$, one straightforward approach of the element-wise uncertainty quantification is conducting the Monte Carlo test with the given statistical distribution of the observation $\mathbf{W}$. However, the size of HSIs is notoriously large, resulting in a heavy computational burden to conduct numerous Monte Carlo tests. Roughly, a one-time process in [20] on a $100 \times 100 \times 100$ HSI requires 20 minutes with a typical commercial desktop computer. Since a robust uncertainty quantification requires a reasonable number of trials, the Monte Carlo test is inefficient for use in the element-wise HSI uncertainty quantification.

Another benefit of this research is that it solves the uncertainty propagation of the sliding-window process. The uniqueness of HSIs is that the permutated 2-D matrix after dimensionality reduction is still relatively large. The solution cannot be achieved with a one-time LRMA process. Instead, researchers have adopted the sliding-window strategy to segment the HSI into many small patches and process them parallel. The different statistical properties of each overlapped patch make it challenging to quantify the uncertainty of the final restored HSI. We solve this problem and present a closed-form solution in the proposed sliding-window element-wise closed-form HSI uncertainty quantification.

B. Uncertainty Quantification for the Exact Low-Rank Factorization

This section focuses on solving the uncertainty quantification of the patch-wise low-rank factorization process (3). We follow the example of [31] and draw a similar conclusion for the uncertainty quantification of the exact low-rank patch factorization. A concise and clear proof is also provided. HSI patch-based averaging (5) will be discussed in Section III-C. A recent study [31] proposed an uncertainty quantification method for the exact low-rank 2-D matrix. This element-wise algorithm can quantify the confidence interval for a common low-rank-based 2-D matrix denoising algorithm. Synthetic and real-world tests validate the efficiency of this method. We introduce this approach and modify it to enable the quantification of the uncertainty of each estimated 3-D HSI patch $\hat{\mathbf{W}}^{(ijk)}$ using (3). We also provide concise proof different from [31].

In the patch optimization process shown in (3), we first aim to quantify the uncertainty of the temporary 2-D square matrix $\hat{\mathbf{W}}^{\sharp(ijk)}$ ($\hat{\mathbf{W}}^{\sharp(ijk)} = g(\hat{\mathbf{W}}^{(ijk)})$ and it can be any patch in Ω). We denote the ground truth of $\hat{\mathbf{W}}^{\sharp(ijk)}$ and $\hat{\mathbf{W}}^{(ijk)}$ as $\mathbf{W}^{\sharp\star(ijk)}$ and $\mathbf{W}^{\star(ijk)}$. For the sake of conciseness, the rank-r SVD decomposition is $\mathbf{W}^{\sharp\star(ijk)} = \mathbf{U}^{\star(ijk)}\Sigma^{\star(ijk)}\mathbf{V}^{\star(ijk)\top}$. For conciseness, we simplify $(\mathbf{U}^{\star(ijk)}, \Sigma^{\star(ijk)}, \mathbf{V}^{\star(ijk)})$ by ignoring the patch index (ijk) and rewrite it as $(\mathbf{U}^{\star}, \Sigma^{\star}, \mathbf{V}^{\star})$. Two matrices are defined as the product of the unitary matrix and the squared singular matrix is: $\mathbf{X}^{\star} \triangleq \mathbf{U}^{\star}\Sigma^{\star 1/2} \in \mathbb{R}^{K \times r}$, $\mathbf{Y}^{\star} \triangleq \mathbf{V}^{\star}\Sigma^{\star 1/2} \in \mathbb{R}^{L \times r}$. The following rules apply:

$$\mathbf{X}^{\star\top}\mathbf{X}^{\star} = \mathbf{Y}^{\star\top}\mathbf{Y}^{\star} = \Sigma^{\star}. \tag{6}$$

Following the above definition, the ground truth 2-D temporary matrix is $\mathbf{W}^{\sharp\star(ijk)} = \mathbf{X}^{\star}\mathbf{Y}^{\star\top}$. Similarly, we define the estimated version $\hat{\mathbf{W}}^{\sharp(ijk)} \approx \hat{\mathbf{X}}\hat{\mathbf{Y}}^{\top}$ where $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ are the estimated matrices regarding $\mathbf{X}^{\star}$ and $\mathbf{Y}^{\star}$. We also obtain the estimated version of $\Sigma^{\star}$ as $\hat{\Sigma}$

$$\hat{\Sigma} = \hat{\mathbf{X}}^{\top}\hat{\mathbf{X}} = \hat{\mathbf{Y}}^{\top}\hat{\mathbf{Y}} \approx \Sigma^{\star}. \tag{7}$$

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It should be noticed that the factorization of $\hat{\mathbf{W}}$ is not unique. For each pair of estimated auxiliary matrices $(\hat{\mathbf{X}}, \hat{\mathbf{Y}})$, there are infinite unitary matrices $\mathbf{T}$ that satisfy

$$\hat{\mathbf{W}}^{\sharp(ijk)} \approx \hat{\mathbf{X}}\hat{\mathbf{Y}}^{\top} = \left(\hat{\mathbf{X}}\mathbf{T}\right)\left(\mathbf{T}^{\top}\hat{\mathbf{Y}}^{\top}\right). \tag{8}$$

Among the unitary matrices, there is a rectification matrix $\hat{\mathbf{R}}$ that minimizes the distance between the auxilliary matrices $(\hat{\mathbf{X}}, \hat{\mathbf{Y}})$ and the ground truth $(\mathbf{X}^{\star}, \mathbf{Y}^{\star})$

$$\min_{\hat{\mathbf{R}}} ||\mathbf{X}\hat{\mathbf{R}} - \mathbf{X}^{\star}||_{\mathbf{F}}^2 + ||\mathbf{Y}\hat{\mathbf{R}} - \mathbf{Y}^{\star}||_{\mathbf{F}}^2. \tag{9}$$

Theorem 1: On condition that each pixel of the observed image $\mathbf{W}$ *only has i.i.d. Gaussian noise* $[\mathbf{G}^{(ijk)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_0^2)$ *in (4)] and the low-rank solver achieves the global minimum, the differences between the rectified auxiliary matrices estimations* $\hat{\mathbf{X}}$ *and* $\hat{\mathbf{Y}}$ *and the ground truth* $\mathbf{X}^{\star}$ *and* $\mathbf{Y}^{\star}$ *are*

$$\begin{aligned} \hat{\mathbf{X}}\hat{\mathbf{R}} - \mathbf{X}^{\star} &= \mathbf{Z}_{\mathbf{X}} + \Psi_X \\ \hat{\mathbf{Y}}\hat{\mathbf{R}} - \mathbf{Y}^{\star} &= \mathbf{Z}_{\mathbf{Y}} + \Psi_Y \end{aligned} \tag{10}$$

where the unitary matrix $\hat{\mathbf{R}} \in \mathbb{R}^{r\times r}$ $(\hat{\mathbf{R}}\hat{\mathbf{R}}^{\top} = \mathbf{I})$ *is the rectification matrix to rectify* $(\hat{\mathbf{X}}, \hat{\mathbf{Y}})$ *to* $(\mathbf{X}^{\star}, \mathbf{Y}^{\star})$. $\Psi_X \in \mathbb{R}^{K\times r}$, $\Psi_Y \in \mathbb{R}^{L\times r}$ *are the negligible residuals. The rows of the error matrix* $\mathbf{Z}_X \in \mathbb{R}^{K\times r}$ *(resp.* $\mathbf{Z}_Y \in \mathbb{R}^{L\times r}$*) are independent and obey*

$$\begin{aligned} \mathbf{Z}_X^{\top}\mathbf{e}_{\upsilon} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma_0^2\left(\Sigma^{\star}\right)^{-1}\right), \quad \text{for } 1 \le j \le K \\ \mathbf{Z}_Y^{\top}\mathbf{e}_{\upsilon} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma_0^2\left(\Sigma^{\star}\right)^{-1}\right), \quad \text{for } 1 \le j \le K \end{aligned} \tag{11}$$

where $\mathbf{e}_{\upsilon}$ *is the basis vector.*

We provide the proof of **theorem** 1 in Section . This proof is specifically designed for the full HSI patch denoising scenario and is clearer and more concise than the one in [31].

In **theorem** 1, we make two assumptions, i.e., the Gaussian noise distribution and the global minimum obtained from the low-rank solver. The former ensures the correctness of the error propagation equations. The latter ensures that the residual noises Ψ_X and Ψ_Y are negligible compared to $\mathbf{Z}_{\mathbf{X}}$ and $\mathbf{Z}_{\mathbf{Y}}$, and (10) is approximated to

$$\begin{aligned} \hat{\mathbf{X}}\hat{\mathbf{R}} - \mathbf{X}^{\star} &\approx \mathbf{Z}_{\mathbf{X}} \\ \hat{\mathbf{Y}}\hat{\mathbf{R}} - \mathbf{Y}^{\star} &\approx \mathbf{Z}_{\mathbf{Y}}. \end{aligned} \tag{12}$$

Assuming that the first-order expansion is reasonably tight and omitting the high order expansion, the element-wise error between the estimation $\hat{\mathbf{W}}_{u\upsilon}^{\sharp(ijk)}$ and the ground truth $\mathbf{W}_{u\upsilon}^{\sharp\star(ijk)}$ at the position $u\upsilon$ is

$$\begin{aligned} &\hat{\mathbf{W}}_{u\upsilon}^{\sharp(ijk)} - \mathbf{W}_{u\upsilon}^{\sharp\star(ijk)} \\ &= \left[\mathbf{e}_u^{\top}\hat{\mathbf{X}}\hat{\mathbf{R}}\left(\hat{\mathbf{Y}}\hat{\mathbf{R}}\right)^{\top}\mathbf{e}_{\upsilon} - \mathbf{e}_u^{\top}\left(\mathbf{X}^{\star}\mathbf{Y}^{\star\top}\right)\mathbf{e}_{\upsilon}\right] \\ &= \left[\mathbf{e}_u^{\top}\left(\hat{\mathbf{X}}\hat{\mathbf{R}} - \mathbf{X}^{\star} + \mathbf{X}^{\star}\right)\left(\hat{\mathbf{Y}}\hat{\mathbf{R}} - \mathbf{Y}^{\star} + \mathbf{Y}^{\star}\right)^{\top}\mathbf{e}_{\upsilon} - \mathbf{e}_u^{\top}\left(\mathbf{X}^{\star}\mathbf{Y}^{\star\top}\right)\mathbf{e}_{\upsilon}\right] \\ &\approx \left[\mathbf{e}_u^{\top}\left(\hat{\mathbf{X}}\hat{\mathbf{R}} - \mathbf{X}^{\star}\right)\mathbf{Y}^{\star\top}\mathbf{e}_{\upsilon} + \mathbf{e}_u^{\top}\mathbf{X}^{\star}\left(\hat{\mathbf{Y}}\hat{\mathbf{R}} - \mathbf{Y}^{\star}\right)^{\top}\mathbf{e}_{\upsilon}\right] \\ &= \left[\mathbf{e}_u^{\top}\left(\mathbf{Z}_X\mathbf{Y}^{\star\top}\right)\mathbf{e}_{\upsilon} + \mathbf{e}_u^{\top}\left(\mathbf{X}^{\star}\mathbf{Z}_Y^{\top}\right)\mathbf{e}_{\upsilon}\right] \end{aligned} \tag{13}$$

where the bases $\mathbf{e}_u$, $\mathbf{e}_{\upsilon}$ localize the elements involved in calculating the element in position (u, υ). After some manipulation, we have the element-wise variance of the error as

$$\begin{aligned} &\operatorname{var}\left(\hat{\mathbf{W}}_{u\upsilon}^{\sharp(ijk)} - \mathbf{W}_{u\upsilon}^{\sharp\star(ijk)}\right) \\ &\overset{\text{(i)}}{\approx} \left[\operatorname{var}\left(\mathbf{e}_u^{\top}\left(\mathbf{Z}_X\mathbf{Y}^{\star\top}\right)\mathbf{e}_{\upsilon}\right) + \operatorname{var}\left(\mathbf{e}_u^{\top}\left(\mathbf{X}^{\star}\mathbf{Z}_Y^{\top}\right)\mathbf{e}_{\upsilon}\right)\right] \\ &\overset{\text{(ii)}}{=} \sigma_0^2\left[\mathbf{e}_{\upsilon}{}^{\top}\mathbf{Y}^{\star}\left(\Sigma^{\star}\right)^{-1}\mathbf{Y}^{\star\top}\mathbf{e}_{\upsilon} + \mathbf{e}_u^{\top}\mathbf{X}^{\star}\left(\Sigma^{\star}\right)^{-1}\mathbf{X}^{\star\top}\mathbf{e}_u\right] \\ &\overset{\text{(iii)}}{=} \sigma_0^2\left(\|\mathbf{U}_{u,\cdot}^{\star}\|_2^2 + \|\mathbf{V}_{\upsilon,\cdot}^{\star}\|_2^2\right) = \sigma_0^2 \upsilon_{u\upsilon}^{\star} \end{aligned} \tag{14}$$

where (i) is derived from **theorem** 1 since $\mathbf{Z}_X$ and $\mathbf{Z}_Y$ are nearly independent. (ii) is obtained from (10). (iii) $\mathbf{U}_{u,\cdot}^{\star}$ and $\mathbf{V}_{\upsilon,\cdot}^{\star}$ are the u/υth row of $\mathbf{U}^{\star}$ and $\mathbf{V}^{\star}$. For conciseness, we define $\upsilon_{u\upsilon}^{\star} = (\|\mathbf{U}_{u,\cdot}^{\star}\|_2^2 + \|\mathbf{V}_{\upsilon,\cdot}^{\star}\|_2^2)$, and $\operatorname{var}(\cdot)$ is the variance.

In practice, the rows of the ground truth $\mathbf{U}_{u,\cdot}^{\star}$ and $\mathbf{V}_{\upsilon,\cdot}^{\star}$ are impossible to obtain. Assuming that the solver achieves a solution close to the global minimum, we approximate them with the estimated version $\hat{\mathbf{U}}_{u,\cdot}$ and $\hat{\mathbf{V}}_{\upsilon,\cdot}$, which are the u/υth row of the estimations $\hat{\mathbf{U}}$ and $\hat{\mathbf{V}}$. Thus the element-wise variance can be approximated as

$$\operatorname{var}\left(\hat{\mathbf{W}}_{u\upsilon}^{\sharp(ijk)} - \mathbf{W}_{u\upsilon}^{\sharp\star(ijk)}\right) \approx \sigma_0^2\left(\|\hat{\mathbf{U}}_{u,\cdot}\|_2^2 + \|\hat{\mathbf{V}}_{\upsilon,\cdot}\|_2^2\right) = \sigma_0^2\hat{\upsilon}_{u\upsilon} \tag{15}$$

where $\hat{\upsilon}_{u\upsilon}$ can be regarded as the estimated version of $\upsilon_{u\upsilon}^{\star}$. Due to the relationship of $\mathbf{W}_{u\upsilon}^{\sharp\star(ijk)}$ and $\hat{\mathbf{W}}_{u\upsilon}^{\sharp(ijk)}$, the corresponding variance of $\hat{\mathbf{W}}_{u\upsilon}^{(ijk)}$ is (16) and provides a closed-form variance describing the confidence of $\hat{\mathbf{W}}_{u\upsilon}^{(ijk)}$. The variance of $\hat{\mathbf{W}}_{u\upsilon}^{(ijk)}$ is

$$\sigma_{\xi}^{(ijk)^2} = \operatorname{var}\left(\hat{\mathbf{W}}_{u\upsilon}^{(ijk)} - \mathbf{W}_{u\upsilon}^{\star(ijk)}\right) \approx g^{-1}\left(\sigma_0^2\hat{\upsilon}_{u\upsilon}\right) \tag{16}$$

where $g^{-1}(\cdot)$ is the inverse permutation function of $g(\cdot)$. $\xi \subseteq \{1, \ldots, M\}\times\{1, \ldots, N\}\times\{1, \ldots, P\}$ is the corresponding 3-D index of the 2-D $u\upsilon$ at patch (ijk).

C. Uncertainty Estimation of the Sliding-Window

Another unique characteristic of the uncertainty quantification in the LRMA technique is the "uncertainty mixing" caused by the sliding-window strategy. The sliding window approach is seldom used in the 2-D image domain, and recent work [31] did not address this issue. Thus, we propose a method to solve the "uncertainty mixing" issue in HSI denoising due to the sliding-window strategy. Unlike averaging all patches to generate the final restored HSI (2), the uncertainty of the product does not follow the typical error propagation, such as estimating the uncertainty of multiple observations [35]. In the scenario of uncertainty estimation for i.i.d. observations, if the variance of each observation is σ_o^2 and we conduct n times observation, the variance of the averaged observation is $(1/n)\sigma_o^2$. This shows that the accuracy of the average observation is n times more accurate than every single observation. However, in the LRMA process, we only make one observation. If we can process the entire HSI using one-time LRMA and the accuracy is σ_o^2, the sliding-window strategy in LRMA should not exceed σ_o^2 because no new information has been introduced. In other words, $(1/n)\sigma_o^2$ is not the correct uncertainty. The difference between multiple i.i.d. observations and the single processing scenarios is that the former has i.i.d. observations for all events, whereas the observations in the latter are correlated. In other words, the covariance between the overlapped patches in the LRMA is

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not null. For two related events X and Y, the ratio η is often used to describe the relationship

$$\eta = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \cdot \text{var}(Y)}} \tag{17}$$

where $\text{cov}(\cdot)$ defines the covariance of the two events. If η is 0, the events X and Y are i.i.d. If η is 1, the events X and Y are linearly correlated, having the highest correlation.

We provide a toy example as the weighted average of events X and Y as $F = aX + bY$. Assuming the variance of X and Y are σ_X^2 and σ_Y^2, the propagated uncertainty of F is σ_F^2

$$\sigma_F^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2\eta_{XY}ab\sigma_X\sigma_Y \tag{18}$$

where η_{XY} is the variance ratio defining the relation between events X and Y.

For convenience, we rewrite (5) to the element-wise form

$$\hat{\mathbf{W}}_\xi = \frac{1}{\phi(\xi)} \sum_{(i,j,k)\in\Phi(\xi)} f_p(\hat{\mathbf{W}}^{(ijk)})_{uv} \quad \text{such that } (u, v) = G(\xi) \tag{19}$$

where $\Phi(\xi)$ defines the index of the patches covering the position (u, v) and $\phi(\xi)$ is the number of those patches. Similar to the permutation function $g(\cdot)$, $G(\cdot)$ converts the 3-D index of $\hat{\mathbf{W}}$ to the 2-D premutated version $\hat{\mathbf{W}}^\sharp$. By denoting the element-wise variance of each $f_p(\hat{\mathbf{W}}^{(ijk)})_{uv}$ is $\sigma_{uv}^{(ijk)}$, based on the uncertainty propagation theory, we obtain the variance that describes $\hat{\mathbf{W}}_{uv}$ as

$$\hat{\sigma}_\xi^2 = \frac{1}{\phi(\xi)^2} \sum_{(i,j,k)\in\phi(\xi)} \sigma_\xi^{(ijk)^2} + \frac{1}{\phi(\xi)^2} \times \sum_{(i,j,k)\in\phi(\xi)} \sum_{(i',j',k')\in\phi(\xi)} 2\eta^{(ijk)(i'j'k')}\sigma_\xi^{(ijk)}\sigma_\xi^{(i'j'k')} \tag{20}$$

where each (i, j, k) is different from (i', j', k'). $\eta^{(ijk)(i'j'k')}$ defines the relation between the element ξ in patch (ijk) and patch $(i'j'k')$. Readers may be aware that this is an extended multiple event version of the toy example in (18).

We provide a practical solution to estimate the optimal ratio $\eta^{(ijk)(i'j'k')}$. Similar to **theorem** 1, we assume that the adopted low-rank solver (GoDec algorithm in this article, but it could be any solver) achieves the global minimum, the information on the denoised element ξ in the patch $\hat{\mathbf{W}}^{(ijk)}$ is determined by the sum of the information embedded in all the elements in the patch $\hat{\mathbf{W}}^{(ijk)}$. The mutual information encoded in two overlapping patches is linearly related to the number of elements shared by the overlapping patches [36]. Thus, the mutual information of the element from two patches is linearly related to the number of shared elements in the two patches. Therefore, the optimal ratio between the two patches $\eta^{(ijk)(i'j'k')}$ is close to

$$\eta^{(ijk)(i'j'k')} \to \left[\frac{N((ijk), (i'j'k'))}{KL}\right]^- \tag{21}$$

where $N((ijk), (i'j'k'))$ defines the number of shared elements in patch (ijk) and patch $(i'j'k')$. $[\cdot]^-$ means slightly below $\cdot$. The ratio on the left is close to but smaller than that of the right term. The result is attributed to the assumption of a global minimum. If the global minimum is guaranteed, we have the information in each patch fully exploited; thus, the ratio is linearly related to the overlap. Thus, the mutual information and number of elements are not strictly linearly related. Extreme case is the full overlap ($\eta^{(ijk)(i'j'k')} = 1$) and no overlap ($\eta^{(ijk)(i'j'k')} = 0$). Otherwise, if Ψ_X and Ψ_Y are not ignored, or the low-rank solver does not converge adequately, the variance based on the local minimum is not accurate and behaves more randomly; thus, the ratio is less than the expected right term. In the extreme case when the low-rank solver fails and converges randomly either due to bad performance of the algorithm or heavy random noise, the $\eta^{(ijk)(i'j'k')}$ should be 0. All other things being equal, a smaller ratio will lead to a smaller variance and slight over-confidence. However, a quantitative assessment of the approximation of (21) is difficult because the performance of the low-rank solver remains difficult to quantify.

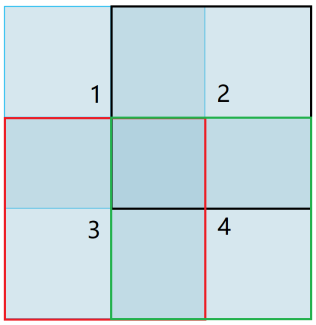


Fig. 3. Example to explain the definition of mutual information defined in (21). Each of the four 2-D overlapping patches 1, 2, 3, and 4 has four pixels.

Fig. 3 presents an example of (21). Each neighboring block pair has 50% overlap. Based on the statistics of the number of overlapping elements, the η of the pairs $(1, 2)$, $(2, 3)$, $(3, 4)$ and $(4, 1)$ is 0.5, whereas the η of pairs $(1, 3)$ and $(2, 4)$ is 0.25.

D. Impact of Random Noise

The result of (16) is based on the assumption that the noises Ψ_X and Ψ_Y are significantly smaller than the variables $\mathbf{Z_X}$ and $\mathbf{Z_Y}$ with a normal distribution. If Ψ_X and Ψ_Y are not ignored, the conclusion does not hold. Therefore, it remains challenging to quantify the impact of random noise versus Gaussian noise. However, as [33] pointed out, the RPCA-like formulation (3) is robust against random noise since the low-rank solver will adjust the use of the norm-1 or norm-2 to regularize the state matrix. Correspondingly, elements not corrupted by random noise are unaffected, whereas the impulse noise can be identified. Therefore, regarding the impact of random noise, the robustness of our approach is closely related to the choice of the LRMA algorithm. It should be noted that most existing HSI denoising studies have focused on modeling Gaussian noise [21], [25]. A few LRMA-based methods have addressed non-Gaussian noise [24]. These approaches utilize a total-variance-related term to filter out non-Gaussian noise in the HSI to maintain good boundaries after filtering. Similarly, the proposed uncertainty estimation method can also be integrated

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with the method in [24] since the total-variance term is only a constraint.

IV. Results and Discussion

Our work aims at providing an uncertainty quantification approach and does not consider convergence analysis, hyperparameter definition, denoised image visualization, or comparison of HSI denoising methods. For clarifications on the methods described in Section II, we refer the reader to [20], [21], [24], [26]–[28]. This work is NOT a new LRMA formulation. We only demonstrate that the proposed closed-form uncertainty estimation approach (Section III) quantifies the denoised HSI accurately, efficiently, and robustly. Since previous works have not provided any uncertainty quantification methods, we only show that our proposed approach can statistically describe the uncertainty based on the Monte Carlo tests.

To the best of our knowledge, a Monte Carlo test is the only appropriate method to validate the efficiency of the proposed method. The complexity of the LRMA formulation hinders the closed-form retrieval of the posterior distribution of the denoised image. This problem motivated us to provide the closed-form estimation of the variance. Therefore, we could not design an experiment providing the ground truth information on both the noisy image and the denoised image. Moreover, regarding the Monte Carlo tests, we should emphasize that the noise of the original image follows identical Gaussian distribution while the noise of the denoised image is nonidentical. The noise of any pixel in original observations follows $\mathbf{G}^{(ijk)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_0^2)$. However, as (16) indicates, the noise of the denoised image depends on the ratio $\hat{o}_{uv}$. Thus, the pixels of the denoised image do not maintain the same noise level. The simulated identical noise can only be enforced on the original image. The Monte Carlo test is the optimal method to obtain the posterior distribution of the denoised image. Theoretically, the Monte Carlo test can generate the ground truth variance when the number of trials is large enough.

We validate the proposed method on the Indian Pines data set (simulation data set),[1] the University of Pavia data set (real data set),[2] and the Washington, D.C. mall data set [3] (real data set). The Indian Pines data set is a subset of the data acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor. It consists of 145×145 pixels and 224 spectral reflectance bands in the wavelength range of $0.4 - 2.510^{-6}$ m and covers the Indian Pines test site in northwestern Indiana, USA. The Indian Pines scene consists of two-thirds agricultural land and one-third forest or other natural perennial vegetation. We use the ground truth of the Indian Pines data set and simulate arbitrary Gaussian noise. The University of Pavia data set was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor and covers Pavia in northern Italy. It consists of 610×610 pixels with 103 spectral bands; however, some of the samples in the images contained no information and were discarded before the analysis. The geometric resolution is 1.3 meters. We select a subset of $610 \times 610 \times 103$ in the experiments. In the Washington, D.C. mall data set, the image contains 191 spectral bands with a size of 1208×307. We clip the image to a size of $256 \times 256 \times 191$ in our experiment. Fig. 4 shows the false-color images of the three data sets.

TABLE I

Coverage Rates of $\mathrm{var}(\hat{\mathbf{W}}_\xi^{\{l\}} - \overline{\hat{\mathbf{W}}}_\xi)$ for Different σ_0 for 100 Monte Carlo Trials. "Std" Denotes the Standard Deviation. The Proposed Closed-Form Uncertainty Estimation Approach Describes the Probability Distribution Well

	Indian		University of Pavia		Washington, D.C.	
σ_0	Mean	Std	Mean	Std	Mean	Std
0.025	0.9514	0.0483	0.9573	0.0515	0.9411	0.0583
0.050	0.9579	0.0371	0.9476	0.0292	0.9368	0.0348
0.075	0.9576	0.0383	0.9556	0.0263	0.9501	0.0311
0.100	0.9643	0.0319	0.9652	0.0233	0.9626	0.0266
0.125	0.9704	0.0273	0.9735	0.0205	0.9712	0.0297

We strictly follow the parameter and noise setting of the selected LRMA method to provide a convincing validation [20]. The size of the sliding window is $20 \times 20 \times P$, where P is the number of bands in the data set. The step size is 4, and the rank r is 7. In the image restoration process, all images are normalized to the range [0 1]. The zero-mean Gaussian noise is uniformly applied to all pixels in the data sets with different levels of noise. Since the two real data sets do not have ground truth, we use the output of [20] as the ground truth. The impact of impulse noise is further tested on the data sets. It should be noted that this research is based on the known raw HSI noise, which can be retrieved using previously described methods [29], [30].

A. Validation of the Uncertainty Description

The accuracy of the proposed closed-form uncertainty quantification method is extensively validated with numerical Monte Carlo tests. We apply different levels of Gaussian noises to the HSI. The numerical experiments provide sufficient samples to test if the statistical distributions of the samples are described accurately with the output confidence interval. Specifically, $T = 100$ Monte Carlo tests are performed on the input of the same noise σ_0^2. We should emphasize that the 100 trials are chosen due to limited computational resources. Due to the large volume of the HSIs, 100 Monte Carlo trials are sufficient to obtain a statistic assessment of the efficiency of the proposed method. The code is provided for readers to conduct more tests.[4] For each element of the estimated HSI in trial l at position ξ as $\hat{\mathbf{W}}_\xi^{\{l\}}$, we define its deviation from the corresponding T times mean value as $\overline{\hat{\mathbf{W}}}_\xi$. The deviation from the mean is defined as

$$e_\xi^{\{l\}} = ||\hat{\mathbf{W}}_\xi^{\{l\}} - \overline{\hat{\mathbf{W}}}_\xi||_1, \quad l \in [1, \ldots, T]. \tag{22}$$

If the closed-form solution describes the statistical distribution of the estimation well, 95% of $\hat{\mathbf{W}}_\xi^{\{l\}}$ should fall within the range of $[\overline{\hat{\mathbf{W}}}_\xi - 1.96\hat{\sigma}_\xi,\ \overline{\hat{\mathbf{W}}}_\xi + 1.96\hat{\sigma}_\xi]$. Following equation (22), we define the ratio of $e_\xi^{\{l\}}$ falling within the $1.96\hat{\sigma}_\xi$ bound as the element-wise **coverage rate**. The average coverage rate of

[1] https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

[2] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[3] https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

[4] https://JingweiSong@bitbucket.org/JingweiSong/uncertainty-estimation-for-hyperspectral-image-denoising.git

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(a) (b) (c)

Fig. 4. (a) False-color image of the Indian Pines data set using bands 10, 37, and 70. This is the manually classified ground truth of the Indian Pines data set. (b) False-color image of the University of Pavia data set composed of the bands 20, 42, and 72. (c) False-color image of the Washington D.C data set composed of bands 60, 27, and 17.

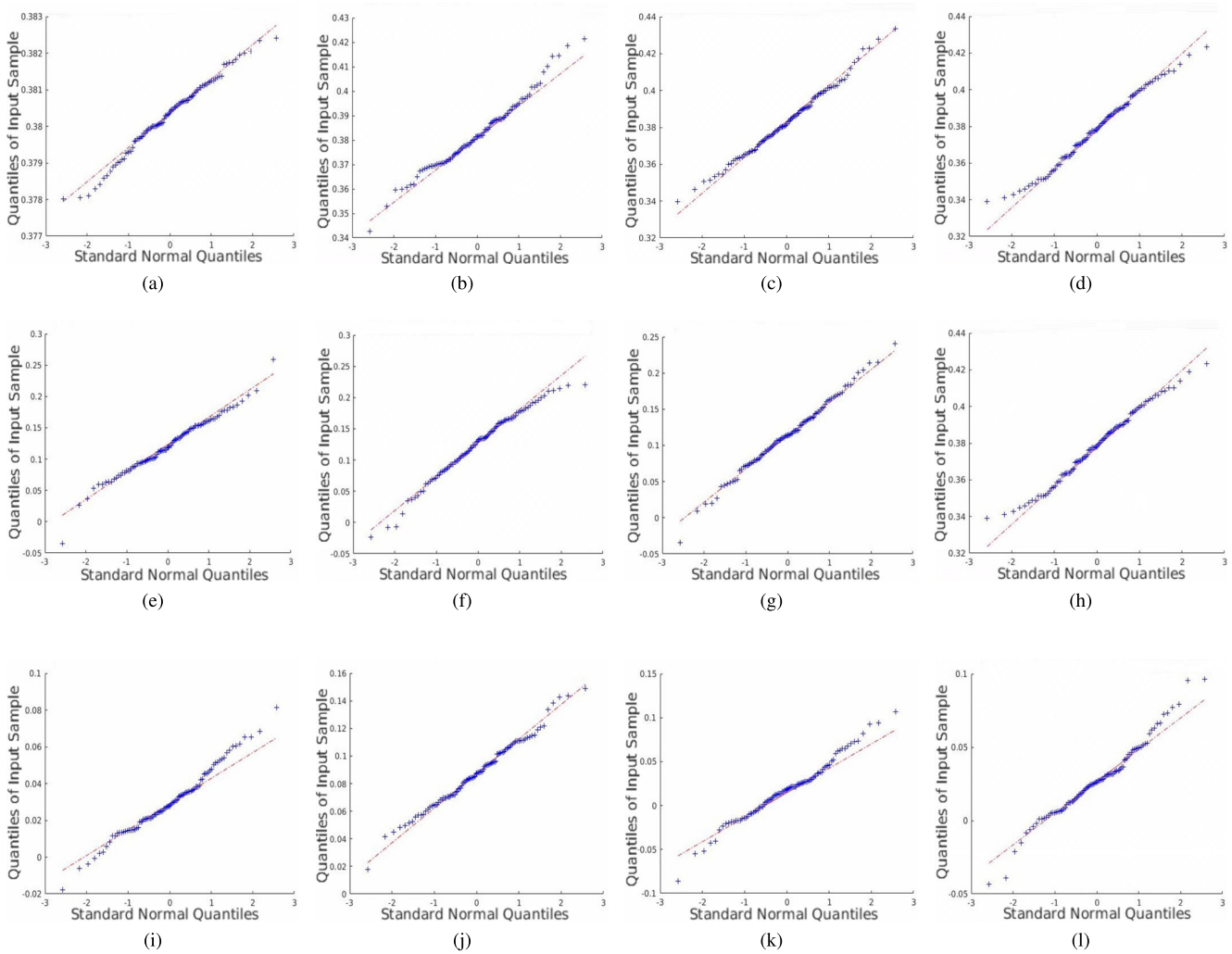


Fig. 5. Q-Q plots of a random element position in 100 independent trials to assess the noise in the three data sets. (a) Indian data set. $\sigma_0 = 0.050$. (b) Indian data set. $\sigma_0 = 0.075$. (c) Indian data set. $\sigma_0 = 0.100$. (d) Indian data set. $\sigma_0 = 0.125$. (e) University of Pavia data set. $\sigma_0 = 0.050$. (f) University of Pavia data set. $\sigma_0 = 0.075$. (g) University of Pavia data set. $\sigma_0 = 0.100$. (h) University of Pavia data set. $\sigma_0 = 0.125$. (i) Washington, D.C. $\sigma_0 = 0.050$. (j) Washington, D.C. $\sigma_0 = 0.075$. (k) Washington, D.C. $\sigma_0 = 0.100$. (l) Washington, D.C. $\sigma_0 = 0.125$.

all elements is the **average coverage rate**. Table I presents the average and standard deviation of the coverage rate. The result indicates that the uncertainty approach robustly and accurately describes the element-wise denoised HSI when the level of noise is low (σ_0 less than 0.100). In the case $\sigma_0 \geq 0.100$, we notice an underconfident of the prediction. We hypothesize that the accuracy is related to the level of noise and the choice of the optimal rank. When the noise level increases, the fixed rank LRMA does not accurately filter the noise while retaining useful information. A lower rank is needed to ensure that the residual follows the assumption of the normal distribution of the noise. As depicted in Table I, we strictly follow [20]

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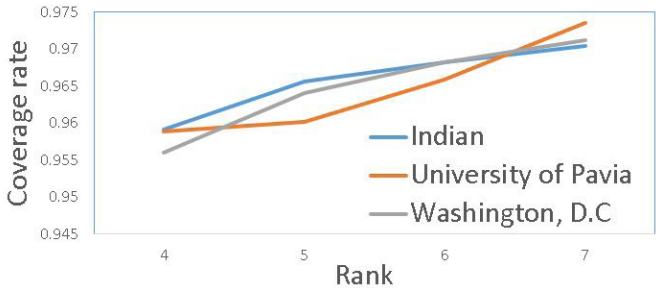


Fig. 6. Relationship between the average coverage rate and the rank at the noise level of $\sigma_0 = 0.125$. As compliment to Table I with fixed scale of noise, this figure provides a comparison of different level of noises.

TABLE II

PROBABILITY VALUES (p-VALUE) OF THE SHAPIRO–WILK TESTS OF THE INDIAN PINES, UNIVERSITY OF PAVIA, AND WASHINGTON, D.C. MALL DATA SETS. WE CITE THE CORRESPONDING ELEMENT POSITIONS SHOWN IN FIG. (5) AS AN EXAMPLE THROUGH ALL THE TESTS. THE p-VALUE > 0.05 INDICATES A SIGNIFICANT POSSIBILITY THAT THE ELEMENT IS NORMALLY DISTRIBUTED

Dataset	Indian	University of Pavia	Washington, D.C.
0.050	0.4239	0.1531	0.2046
0.075	0.1103	0.1924	0.0585
0.100	0.8498	0.7537	0.6977
0.125	0.4346	0.1531	0.5331

by setting $r = 7$. To determine the reason, we conduct an experiment by choosing different r. The results are shown in Fig. (6), indicating that the performance (uncertainty quantification) of our approach improves by decreasing the arbitrary rank. This implies that the Monte Carlo experiment provides a method to validate the correct choice of the rank. Further studies are needed to determine the rank at a high noise level.

B. Validation of the Normal Distribution of the Restored HSI

In addition to the **average coverage rate**, we also adopt the Monte Carlo experiments to prove that the restored HSI $\hat{\mathbf{W}}^{(l)}$ has an innate Gaussian distribution. Quantile–Quantile (Q–Q) plots are created, and Shapiro–Wilk tests are conducted to validate the proposed approach. Fig. 5 shows the Q-Q plots of all the data sets. Note that we randomly plot one element in each data set as an example, but all other elements are similar since the observation is isotropic. Fig. 5 indicates that the Monte Carlo estimation of all the trials $\hat{\mathbf{W}}^{(l)}$ is close to a line; thus, the sample follows the Gaussian distribution.

In the Shapiro–Wilk tests, it is not possible to present all the elements. Thus, we randomly choose several elements from the three data sets with different levels of Gaussian noise. The results are presented in Table II. The p-values indicate that all samples follow the normal distribution (> 0.05). Since all pixels are acquired under the same conditions, and their statistics should be consistent (either obey or reject the normal distribution assumption), we can conclude that the restored HSI still follows the Gaussian distribution.

C. Time Consumption Comparison

In addition to the accuracy tests, we determine the time consumption. We find that the closed-form uncertainty estimation method has similar time consumption as the LRMA and far less time consumption than the Monte Carlo numerical tests.

TABLE III

TIME CONSUMPTION OF THE MONTE CARLO UNCERTAINTY ESTIMATION, THE LRMA METHOD [20], AND THE PROPOSED METHOD. NOTE THAT 100 TRIALS ARE CONDUCTED IN THE MONTE CARLO TESTS. THE TIME IS RECORDED IN SECONDS

Dataset	Monte Carlo	LRMA	LRMA+Uncertainty
Indian	7862	653	864
University of Pavia	8393	815	930
Washington, D.C.	12437	1023	1590

We use the LRMA code used in [20] and add the uncertainty estimation module. The proposed algorithm, the LRMA, and the Monte Carlo tests are all implemented in MATLAB 2019a. In the Monte Carlo tests, we utilize the Parallel Computing Toolbox of MATLAB to speed up the steps. The hardware set is 3.1 GHz Intel Xeon E5-2687wv3 and 64 GB DDR4-RAM. In the Monte Carlo experiments, all 20 CPU threads are utilized. Note that in the Monte Carlo tests, each element is covered by several sliding windows. Since the memory is not sufficient to store all sliding windows, we calculate the result of the patch, store it on the hard disk, and reload it to calculate each element in the final step.

Table III shows the time consumption of the Monte Carlo uncertainty estimation, the LRMA method [20], and the proposed method integrated into [20]. The 100 Monte Carlo trials are significantly speeded up because of the running in parallel using 20 CPU threads. The results indicate that the proposed closed-form estimation does not add a significant amount of time to the LRMA method, whereas the Monte Carlo test requires substantially more time. As indicated by (14), the proposed approach requires no more than one band-wise SVD decomposition of the given HSI in addition to the original LRMA solver.

D. Impact of Random Noise

In addition to the coverage rate tests, we assess if the proposed approach is robust to the random noise from the sensor. As (2) shows, our work ignores the random noise $\mathbf{N}$ and only assumes that the residuals follow the Gaussian noise $\mathbf{G}$. As described in Section III-D, we expect that the proposed method is robust to random noise such as impulse noise, deadlines, and stripes [37], [38]. We conduct the tests by deliberately corrupting the original HSI with random noise and test the performance of the proposed uncertainty quantification method. We simulate different proportions of impulse noise on all data sets and determine the coverage rate of the Monte Carlo experiments.

Table IV shows the results of the Monte Carlo experiments for different coverage rates. The results indicate a relationship between the robustness of the proposed closed-form method and the coverage rate. When the noise σ_0 is greater than 0.1, the uncertainty quantification approach is less sensitive to pulse noises. When the noise σ_0 is less than 0.075, the proposed method is susceptible to random noise. Theoretically, the proposed approach is only effective for Gaussian noise, which is the low-rank prior to (2). Therefore, less Gaussian noise means less reliable of the assumption that $\mathbf{N}$ is less than $\mathbf{G}$ in (1). Thus, the results from the closed-form approach are less reliable. When the Gaussian noise level of the observation

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TABLE IV

COVERAGE RATE AND STANDARD DEVIATION OF DIFFERENT RATIOS OF IMPULSE NOISE. WE ADD $m\%$, $m \in [5, 10, 20, 30]$ RATIOS OF IMPULSE NOISE TO THE OBSERVATIONS, AND TEST THE COVERAGE RATES OF DIFFERENT SCENARIOS IN 100 MONTE CARLO TRIALS

		5%		10%		20%		30%	
Dataset	σ_0	Mean	Std	Mean	Std	Mean	Std	Mean	Std
Indian	0.050	0.9006	0.0619	0.8429	0.0770	0.7309	0.0924	0.6278	0.0952
	0.075	0.9348	0.0476	0.9063	0.0578	0.8395	0.0745	0.7621	0.0860
	0.100	0.9516	0.0382	0.9340	0.0456	0.8888	0.0606	0.8293	0.0747
	0.125	0.9624	0.0318	0.9495	0.0379	0.9148	0.0516	0.8657	0.0663
University of Pavia	0.050	0.9008	0.0438	0.8472	0.0558	0.7466	0.0711	0.6578	0.0783
	0.075	0.9434	0.0343	0.9189	0.0432	0.8640	0.0591	0.8020	0.0715
	0.100	0.9649	0.0275	0.9519	0.0340	0.9195	0.0477	0.8778	0.0607
	0.125	0.9778	0.0220	0.9700	0.0270	0.9496	0.0382	0.9201	0.0503
Washington, D.C	0.050	0.9010	0.0549	0.8536	0.0690	0.7649	0.0860	0.6840	0.0933
	0.075	0.9456	0.0417	0.9250	0.0514	0.8781	0.0689	0.8248	0.0835
	0.100	0.9674	0.0319	0.9566	0.0388	0.9301	0.0534	0.8954	0.0688
	0.125	0.9798	0.0245	0.9735	0.0297	0.9566	0.0417	0.9330	0.0560

is low, the proposed approach tends to be over-confident of the result, and the result is lower than the coverage rate with a higher σ_0. A higher σ_0 means that the distribution is closer to the impulse noise. Furthermore, since the solvers (either GoDec or ALM solver) are all based on SVD decomposition, our approach has the same computational complexity as the original LRMA methods.

V. CONCLUSION

This is the first study that addresses the uncertainty quantification of LRMA-based HSI denoising techniques. We provide a time-efficient closed-form uncertainty propagation model to quantify the element-wise uncertainty of the denoised HSI without requiring any ground truth. We also propose a weighted average uncertainty formulation to tackle the "uncertainty mixing" problem caused by the sliding-window strategy in HSI denoising. Moreover, the proposed uncertainty estimation approach is independent of the choice of the LRMA solver. It is based on the global minimum assumption of LRMA; thus, a more accurate algorithm leads to a better performance of the uncertainty estimation. Extensive numerical experiments indicate that the proposed uncertainty quantification method is accurate and robust and describes the statistical behavior of the estimates obtained from the LRMA approach. The time consumption of the closed-form formulation is low; therefore, the method can be deployed in any state-of-the-art LRMA algorithm. The proposed uncertainty approach can also be used in applications that require multisource or multiscale data fusion, data assimilation, and remote sensing uncertainty estimation.

Future works may also focus on tackling the observable random noise, such as impulse noises, deadlines, and stripes. The percentage of the observable random noise can be utilized to adjust the proposed closed-form uncertainty method.

APPENDIX

Proof of Theorem 1: If the selected LRMA solver provides the optimal solution to the objective function (3), the optimizer $\hat{\mathbf{X}}$, $\hat{\mathbf{Y}}$ should satisfy the local minima of (3). $F(\hat{\mathbf{X}}, \hat{\mathbf{Y}})$ are denoted as

$$\begin{aligned} F\big(\hat{\mathbf{X}}, \hat{\mathbf{Y}}\big) &= \frac{1}{2}\|\mathbf{W}^{(ijk)} - g^{-1}\big(\hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big)\|_F^2 \\ &\overset{\text{(i)}}{=} \frac{1}{2}\|\mathbf{W}^{\sharp(ijk)} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\|_F^2 \end{aligned} \tag{23}$$

where (i) is because $g^{-1}(\cdot)$ is the permutation function and $g(\mathbf{W}^{(ijk)}) = \mathbf{W}^{\sharp(ijk)}$. Assuming the solver yields $F(\hat{\mathbf{X}}, \hat{\mathbf{Y}})$ close to the global minimum $F(\mathbf{X}^\star, \mathbf{Y}^\star)$. The first-order expansion of $F(\hat{\mathbf{X}}, \hat{\mathbf{Y}})$ is close to the zero matrix defined as $\mathcal{O}$. By solving (1) and ignoring random noise, we define the patch function

$$\begin{aligned} \mathbf{W}^{(ijk)} &= \mathbf{W}^{\star(ijk)} + \mathbf{G}^{(ijk)} \\ \mathbf{W}^{\sharp(ijk)} &= \mathbf{W}^{\sharp\star(ijk)} + \mathbf{G}^{\sharp(ijk)} \end{aligned} \tag{24}$$

where $\mathbf{G}^{\sharp(ijk)} = g(\mathbf{G}^{(ijk)})$. The optimizer $F(\hat{\mathbf{X}}, \hat{\mathbf{Y}})$ achieves local minima which is close to the global minima; thus, we obtain the partial derivatives equal to the zero matrix

$$\begin{aligned} \frac{\partial F\big(\hat{\mathbf{X}}, \hat{\mathbf{Y}}\big)}{\partial \hat{\mathbf{X}}} &\overset{\text{(i)}}{=} \big(\mathbf{W}^{\sharp(ijk)} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big)\hat{\mathbf{Y}} \\ &= \big(\mathbf{G}^{\sharp(ijk)} + \mathbf{W}^{\star(ijk)} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big)\hat{\mathbf{Y}} \\ &= \big(\mathbf{G}^{\sharp(ijk)} + \mathbf{X}^\star\mathbf{Y}^{\star\top} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big)\hat{\mathbf{Y}} \\ &\overset{\text{(ii)}}{\approx} \big(\mathbf{G}^{\sharp(ijk)} + \mathbf{X}^\star\hat{\mathbf{R}}^\top\hat{\mathbf{Y}}^\top - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big)\hat{\mathbf{Y}} \approx \mathcal{O} \end{aligned} \tag{25}$$

$$\begin{aligned} \frac{\partial F\big(\hat{\mathbf{X}}, \hat{\mathbf{Y}}\big)}{\partial \hat{\mathbf{Y}}} &\overset{\text{(i)}}{=} \hat{\mathbf{X}}^\top\big(\mathbf{W}^{\sharp(ijk)} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big) \\ &= \hat{\mathbf{X}}^\top\big(\mathbf{G}^{\sharp(ijk)} + \mathbf{W}^{\star(ijk)} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big) \\ &= \hat{\mathbf{X}}^\top\big(\mathbf{G}^{\sharp(ijk)} + \mathbf{X}^\star\mathbf{Y}^{\star\top} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big) \\ &\overset{\text{(ii)}}{\approx} \hat{\mathbf{X}}^\top\big(\mathbf{G}^{\sharp(ijk)} + \hat{\mathbf{X}}\hat{\mathbf{R}}\mathbf{Y}^{\star\top} - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\big) \approx \mathcal{O} \end{aligned} \tag{26}$$

where (i) is the result of $d\mathbf{W}^{\sharp(ijk)} = d\hat{\mathbf{X}}\hat{\mathbf{Y}}^\top + \hat{\mathbf{X}}(d\hat{\mathbf{Y}})^\top$. (ii) is approximated by the fact that the rectified local minima ($\hat{\mathbf{X}}\hat{\mathbf{R}}$,$\hat{\mathbf{Y}}\hat{\mathbf{R}}$) are close to the global minima ($\mathbf{X}^\star$,$\mathbf{Y}^\star$). We manipulate (25) and (26) to

$$\begin{aligned} \mathbf{G}^{\sharp(ijk)}\hat{\mathbf{Y}} &\approx \big(\hat{\mathbf{X}} - \mathbf{X}^\star\hat{\mathbf{R}}^\top\big)\hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} \\ \big(\hat{\mathbf{X}} - \mathbf{X}^\star\hat{\mathbf{R}}^\top\big) &\approx \mathbf{G}^{\sharp(ijk)}\hat{\mathbf{Y}}\big(\hat{\mathbf{Y}}^\top\hat{\mathbf{Y}}\big)^{-1} \\ \big(\hat{\mathbf{X}}\hat{\mathbf{R}} - \mathbf{X}^\star\big) &\approx \mathbf{G}^{\sharp(ijk)}\hat{\mathbf{Y}}\big(\hat{\mathbf{Y}}^\top\hat{\mathbf{R}}^\top\hat{\mathbf{R}}\hat{\mathbf{Y}}\big)^{-1}\hat{\mathbf{R}} \\ \big(\hat{\mathbf{X}}\hat{\mathbf{R}} - \mathbf{X}^\star\big) &\overset{\text{(i)}}{\approx} \mathbf{G}^{\sharp(ijk)}\hat{\mathbf{Y}}\hat{\mathbf{R}}\big(\hat{\mathbf{Y}}^\top\hat{\mathbf{R}}^\top\hat{\mathbf{R}}\hat{\mathbf{Y}}\big)^{-1} \\ \big(\hat{\mathbf{R}}^\top\hat{\mathbf{X}}^\top - \mathbf{X}^{\star\top}\big)\mathbf{e}_v &\approx \left[\mathbf{G}^{\sharp(ijk)}\hat{\mathbf{Y}}\left(\hat{\mathbf{Y}}^\top\hat{\mathbf{Y}}\right)^{-1}\right]^\top \\ &\quad \times \mathbf{e}_v \overset{\text{(ii)}}{\sim} \mathcal{N}\Big(0, \sigma_0^2\big(\Sigma^\star\big)^{-1}\Big) \\ \hat{\mathbf{X}}^\top\mathbf{G}^{\sharp(ijk)} &= \hat{\mathbf{X}}^\top\hat{\mathbf{X}}\big(\hat{\mathbf{Y}}^\top - \hat{\mathbf{R}}\mathbf{Y}^{\star\top}\big) \\ \big(\hat{\mathbf{Y}}^\top - \hat{\mathbf{R}}\mathbf{Y}^{\star\top}\big) &\approx \big(\hat{\mathbf{X}}^\top\hat{\mathbf{X}}\big)^{-1}\hat{\mathbf{X}}^\top\mathbf{G}^{\sharp(ijk)} \end{aligned} \tag{27}$$

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$$\begin{aligned}\left(\hat{\mathbf{R}}^\top\hat{\mathbf{Y}}^\top-\mathbf{Y}^{\star\top}\right)&\approx\hat{\mathbf{R}}^\top\left(\hat{\mathbf{X}}^\top\hat{\mathbf{R}}^\top\hat{\mathbf{R}}\hat{\mathbf{X}}\right)^{-1}\hat{\mathbf{X}}^\top\mathbf{G}^{\sharp(ijk)}\\ \left(\hat{\mathbf{R}}^\top\hat{\mathbf{Y}}^\top-\mathbf{Y}^{\star\top}\right)&\overset{\text{(i)}}{\approx}\left(\hat{\mathbf{X}}^\top\hat{\mathbf{R}}^\top\hat{\mathbf{R}}\hat{\mathbf{X}}\right)^{-1}\hat{\mathbf{R}}^\top\hat{\mathbf{X}}^\top\mathbf{G}^{\sharp(ijk)}\\ \left(\hat{\mathbf{R}}^\top\hat{\mathbf{Y}}^\top-\mathbf{Y}^{\star\top}\right)\mathbf{e}_v&\approx\left(\widehat{\mathbf{X}}^\top\widehat{\mathbf{X}}\right)^{-1}\widehat{\mathbf{X}}^\top\mathbf{G}^{\sharp(ijk)}\\ &\quad\times\mathbf{e}_v\overset{\text{(iii)}}{\sim}\mathcal{N}\left(0,\sigma_0^2\left(\Sigma^\star\right)^{-1}\right)\end{aligned}\tag{28}$$

where the matrix exchange in step (i) is enabled because $(\hat{\mathbf{X}}^\top\hat{\mathbf{R}}^\top\hat{\mathbf{R}}\hat{\mathbf{X}})$ and $(\hat{\mathbf{Y}}^\top\hat{\mathbf{R}}^\top\hat{\mathbf{R}}\hat{\mathbf{Y}})$ are diagonal matrices. We denote the rectified local minima $(\widehat{\mathbf{X}},\widehat{\mathbf{Y}})$ equal to $(\hat{\mathbf{X}}\hat{\mathbf{R}},\hat{\mathbf{Y}}\hat{\mathbf{R}})$ for a concise presentation.

The proof of step (ii): Each element of the Gaussian noise matrix $\mathbf{G}^{\sharp(ijk)}\sim\mathcal{N}(0,\sigma^2)$. According to (7), $\Sigma^\star$ is a diagonal matrix and the diagonal element at position $[i,i]$ $(\Sigma^\star_{ii})^{-1}$ equals to $1/||\mathbf{X}_{.,i}||_2^2$ or $1/||\mathbf{Y}_{.,i}||_2^2$ where $\widehat{\mathbf{X}}_{.,i}$ or $\widehat{\mathbf{Y}}_{.,i}$ is the ith column of matrix $\widehat{\mathbf{X}}$ or $\widehat{\mathbf{Y}}$.

The proof of (i)

$$\begin{aligned}&\left[\mathbf{G}^{\sharp(ijk)}\widehat{\mathbf{Y}}\left(\widehat{\mathbf{Y}}^\top\widehat{\mathbf{Y}}\right)^{-1}\right]^\top\mathbf{e}_v\\ &\quad=\left[\sum_{i=1}^{L}\frac{1}{\|\widehat{\mathbf{Y}}_{.,1}\|_2^2}\widehat{\mathbf{Y}}_{i1}\mathbf{G}_{vi}^{\sharp(ijk)}\ \cdots\ \sum_{i=1}^{L}\frac{1}{\|\widehat{\mathbf{Y}}_{,r}\|_2^2}\widehat{\mathbf{Y}}_{ir}\mathbf{G}_{vi}^{\sharp(ijk)}\right].\end{aligned}\tag{29}$$

Take the first element $\sum_{i=1}^{L}(1)/(\|\widehat{\mathbf{Y}}_{.,1}\|_2^2)\widehat{\mathbf{Y}}_{i1}\mathbf{G}_{vi}^{\sharp(ijk)}$ as an example

$$\begin{aligned}\operatorname{var}\left(\sum_{i=1}^{L}\frac{1}{\|\widehat{\mathbf{Y}}_{.,1}\|_2^2}\widehat{\mathbf{Y}}_{i1}\mathbf{G}_{vi}^{\sharp(ijk)}\right)&=\left(\sum_{i=1}^{L}\frac{1}{\|\widehat{\mathbf{Y}}_{.,1}\|_2^2}\widehat{\mathbf{Y}}_{i1}\right)^2\sigma_0^2\\ &=\frac{1}{\left(\|\widehat{\mathbf{Y}}_{.,1}\|_2^2\right)^2}\left(\sum_{i=1}^{L}\widehat{\mathbf{Y}}_{i1}\right)^2\sigma_0^2\\ &=\frac{1}{\left(\|\widehat{\mathbf{Y}}_{.,1}\|_2^2\right)}\sigma_0^2\\ &\approx\left(\Sigma^\star_{11}\right)^{-1}\sigma_0^2.\end{aligned}\tag{30}$$

The same conclusion also applies to other elements.

Similarly, the proof of (ii)

$$\begin{aligned}&\left(\widehat{\mathbf{X}}^\top\widehat{\mathbf{X}}\right)^{-1}\widehat{\mathbf{X}}\mathbf{G}^{\sharp(ijk)}\mathbf{e}_v\\ &\quad=\left[\sum_{i=1}^{K}\frac{1}{\|\widehat{\mathbf{X}}_{.,1}\|_2^2}\widehat{\mathbf{X}}_{i1}\mathbf{G}_{iv}^{\sharp(ijk)}\ \cdots\ \sum_{i=1}^{K}\frac{1}{\|\widehat{\mathbf{X}}_{,r}\|_2^2}\widehat{\mathbf{X}}_{i1}\mathbf{G}_{iv}^{\sharp(ijk)}\right].\end{aligned}\tag{31}$$

Take the first element $\sum_{i=1}^{K}(1)/(\|\widehat{\mathbf{X}}_{.,1}\|_2^2)\widehat{\mathbf{X}}_{iv}\mathbf{G}_{i1}^{\sharp(ijk)}$ as an example

$$\begin{aligned}\operatorname{var}\left(\sum_{i=1}^{K}\frac{1}{\|\widehat{\mathbf{X}}_{.,1}\|_2^2}\widehat{\mathbf{X}}_{i1}\mathbf{G}_{iv}^{\sharp(ijk)}\right)&=\left(\sum_{i=1}^{K}\frac{1}{\|\widehat{\mathbf{X}}_{.,1}\|_2^2}\widehat{\mathbf{X}}_{i1}\right)^2\sigma_0^2\\ &=\frac{1}{\left(\|\widehat{\mathbf{X}}_{.,1}\|_2^2\right)^2}\left(\sum_{i=1}^{K}\widehat{\mathbf{X}}_{i1}\right)^2\sigma_0^2\\ &=\frac{1}{\left(\|\widehat{\mathbf{X}}_{.,1}\|_2^2\right)}\sigma_0^2\\ &\approx\left(\Sigma^\star_{11}\right)^{-1}\sigma_0^2.\end{aligned}\tag{32}$$

The same conclusion also applies to other elements.

Following equations (27) and (28), we strictly obtain the conclusion that the residual of Gaussian noise $\mathbf{G}^{(ijk)}\overset{\text{i.i.d.}}{\sim}\mathcal{N}(0,\sigma_0^2)$. Thus, $\Psi_X\in\mathbb{R}^{K\times r}$, $\Psi_Y\in\mathbb{R}^{L\times r}$ are the negligible residuals due to numerical issues.

This concludes the proof of (i) and (ii) and also the proof of **theorem 1**.

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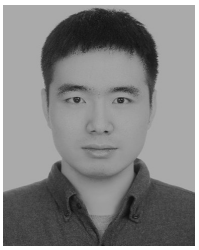
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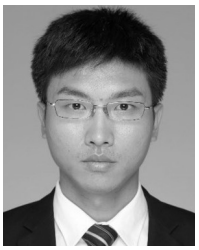
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